Title: Temporal Trends and Mortality in Tracheobronchial Stenting in the United States: a national cohort study

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Abbreviations List:
CAO: central airway obstruction; CPT: Current Procedural Terminology; ICD: International Classification of Diseases; IQR: interquartile range; QoL: quality of life; SASD: State Ambulatory Surgery and Services Databases; SID: State Inpatient Databases

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Authors' contributions:
MTW and HCP were involved in study design and conception. MTW was involved in data acquisition. MTW, JT, CMK, LBY, FM, and HCP performed data analysis. MTW, JT, CMK, LBY, FM, and HCP were involved in data interpretation. MTW drafted the manuscript. MTW, JT, CMK, LBY, FM, and HCP were involved in critical manuscript review. All authors participated in final manuscript revision and take responsibility for the integrity of the data and the accuracy of the data analysis.

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ABSTRACT

Background: Recently published guidelines suggest tracheobronchial stenting for palliating symptomatic central airway obstruction (CAO) that is not amenable to definitive surgical intervention. However, population-based data on the use of tracheobronchial stenting for CAO are lacking.

Methods: This is an observational cohort study of adults who underwent tracheobronchial stenting while hospitalized using the National Inpatient Sample (2016-2022), which represents 97% of all acute care hospitalizations in the US. The primary outcome of interest was the annual incidence of tracheobronchial stenting per 100,000 residents. Separately, to determine the proportion of procedures performed during inpatient hospitalization as opposed to as an ambulatory procedure, we used the State Inpatient Databases and State Ambulatory Surgery and Services Databases across six states in 2019 (which include data for individual states).

Results: Among all hospitalizations in the US from 2016-2022, 0.71 hospitalizations per 100,000 residents received tracheobronchial stenting (95% CI: 0.66, 0.76), increasing from 0.58 in 2016, peaking at 0.77 in 2021, and 0.67 in 2022. Among hospitalized patients who received tracheobronchial stenting, 69.7% had a diagnosis of malignancy and 12.3% died prior to discharge. The proportion of patients undergoing tracheobronchial stenting with a diagnosis of malignancy increased from 65.9% in 2016 to 74.6% in 2022. In hospital mortality among patients receiving tracheobronchial stenting during hospitalization increased from 11.3% in 2016 to 12.7% in 2022. In 2019, across six geographically diverse states in the US, 73.3% of tracheobronchial stenting procedures were performed during inpatient hospitalizations.

Conclusion: Tracheobronchial stenting incidence and mortality for CAO have both increased over time. Regional variation and high in-hospital mortality highlight the need to better define indications and track outcomes of tracheobronchial stenting in patients with CAO.

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Central airway obstruction (CAO) is a devastating condition for patients. Limitation of airflow in the central airways reduces lung function, causes severe symptoms, and contributes to reduced quality of life (QoL) for patients.[1,2] The management strategy is surgical, but many patients with CAO are not surgical candidates due to underlying disease process, overall prognosis, and comorbid conditions.[3–5]

Tracheobronchial stents have therefore been used as a non-surgical alternative to improve lung function and palliate symptoms in patients with CAO.[2,3,6] Most patients with CAO treated with tracheobronchial stenting experience improvement in airway caliber, symptoms, and QoL. Earlier studies investigating the use of tracheobronchial stenting in CAO suggested a survival benefit but a recent randomized controlled trial showed no mortality benefit.[7–11] Recently published guidelines suggest using tracheobronchial stenting for palliating symptomatic central airway obstruction that is not amenable to definitive surgical intervention, but there remains a paucity of data on clinical outcomes.[12–15]

Currently, it is unclear what the incidence of tracheobronchial stenting is in the United States, how this is changing over time, whether stents are predominantly placed during inpatient hospitalization versus as an outpatient procedure (i.e., ambulatory surgery), and how tracheobronchial stent utilization varies across states and regions. An understanding of these critically important parameters is needed to better understand the true patient centered impacts of tracheobronchial stenting.

To understand current practices, we used the National Inpatient Sample (NIS) to assess for the incidence of tracheobronchial stenting performed during inpatient hospitalizations per 100,000 residents in the United States (US), measure trends over time, and assess for regional variation. We separately used six State Inpatient Databases (SID) and State Ambulatory Surgery and Services Databases (SASD) to determine how often airway stenting is performed during inpatient hospitalization versus as an ambulatory surgery and characterize state-level variation. We hypothesized that the majority of tracheobronchial stenting occurs during inpatient hospitalization, that use is increasing over time (that the increase in dedicated Interventional Pulmonology training programs, availability of stents, and ease of placement outweigh any changes from earlier lung cancer diagnosis and improved therapies), and that there is substantial variation across states within the US.

Methods

Data Source

We performed a retrospective cohort study using the NIS from 2016-2022 and the SID and SASD for six states in the United States in 2019 (Florida, Iowa, Maryland, Nebraska, New York, and Wisconsin, selected based on data availability). The National Inpatient Sample (NIS) is a database within the Healthcare Cost and Utilization Project sponsored by the Agency for Healthcare Research and Quality.[16] The NIS is the largest publicly available all-payer inpatient database in the US.[16] It contains a 20% stratified sample of discharges from all hospitals across the US (excluding rehabilitation and long-term acute care hospitals).[16] Each year of data includes over 7 million inpatient stays, which after weighting, represents more than 35 million hospitalizations and accounts for more than 97% of the all acute hospitalizations in the US.[16]

The SID and SASD are separate all-payor databases maintained by the Healthcare Cost and Utilization Project, sponsored by the Agency for Healthcare Research and Quality.[17,18] The SID includes inpatient hospitalizations for individual states, while the SASD includes ambulatory surgeries. We selected 2019, the most recent year with complete data for each state pre COVID-19 pandemic given its impact on healthcare systems and specifically surgical volume.[19–21] We selected the six states of interest because they had both inpatient and outpatient data available.

Study population & primary outcome

Our primary cohort was adult (age ≥18 years) hospitalizations (NIS and SID) and ambulatory procedures (SASD) that received tracheobronchial stenting as identified by *International Classification of Diseases (ICD) 10* procedural codes for inpatient hospitalizations (0B718D*, 0B728D*, 0B738D*, 0B748D*, 0B758D*, 0B768D*, 0B778D*, 0B788D*, 0B798D*, 0B7B8D*, 0BH08D*) and Current Procedural Terminology (CPT) codes for ambulatory surgeries (31631, 31636, 31637).

The primary outcome of interest was the incidence of tracheobronchial stents placed per 100,000 residents using United States Census Bureau national population estimates and state population estimates.[19–23] We identified comorbidities via diagnostic codes during hospitalization using Elixhauser Comorbidity Software.[24–26] To account for illness severity during hospitalizations, we identified acute organ dysfunctions using diagnostic and procedural codes.[27,28]

Subgroup analyses

We performed three subgroup analyses, comparing (1) patients with versus without malignancy; (2) patients who underwent early stenting (performed within 3 days of hospitalization) versus late stenting (performed later than 3 days following admission); and (3) patients who survived vs died during index hospitalization.

Population estimates & trends over time

To assess for change over time, we calculated the incidence of tracheobronchial stenting performed during inpatient hospitalizations in the US per 100,000 residents using sampling weights from NIS, and accounted for stratification and clustering of data.[16,29] We assessed for trends over time comparing the incidence per 100,000 residents. We used state level census to estimate the incidence per 100,000 residents across Census Tract Divisions (as defined by the NIS). We also tested for statistical significance using a non-parametric test for trend over time. However, given the large number of hospitalizations, we do not believe that statistical significance confirms clinical significance.

State level analyses

To determine the proportion of tracheobronchial stenting procedures performed during inpatient hospitalizations versus as an ambulatory procedure, we used the SID and

SASD. Specifically, we compared the incidence of tracheobronchial stenting per 100,000 residents of a state performed during inpatient hospitalizations to the incidence of tracheobronchial stenting per 100,000 residents of a state performed as an ambulatory procedure. We used the incidence across each state to estimate a range performed during inpatient hospitalization as opposed to as an ambulatory procedure. We used the overall incidence per 100,000 to characterize state-level variation.

Data management and analysis were performed in Stata/MP version 19.5 (StataCorp, College Station, TX). We present the incidence of tracheobronchial stenting per 100,000 residents and patient characteristics using standard descriptive statistics. We compared differences between inpatient and ambulatory procedures using ANOVA for continuous variables and chi-square tests for categorical data and considered a $P<0.05$ to be significant. This study follows the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines for observational studies and best practices for working with the SID, SASD, and NIS.[17,18,30] The study was deemed exempt by the University of Michigan IRB (HUM00275375). Data were analyzed from May 2, 2025 to July 25, 2025.

Results

Among all hospitalizations in the United States from 2016-2022, 0.71 hospitalizations per 100,000 residents in the US received tracheobronchial stenting (95% CI: 0.66, 0.76) (**Table 1**). Among patients hospitalized for tracheobronchial stenting, 49.5% were female, the mean age was 62 (95% CI: 62, 63), and 70% had a diagnosis of malignancy

(**Table 1**). Procedures were performed in diverse hospitals including hospital location and teaching status (43.1% in urban teaching hospitals, 30.1% in urban nonteaching hospitals, and 26.8% in rural hospitals) and hospital size (33.5% in large hospitals, 22.9% in medium sized hospitals, and 43.5% in small hospitals). Among patients receiving tracheobronchial stenting, 69.6% were discharged to home (including with home health services) and 12.3% died during hospitalization (**Table 1**).

Subgroup analyses

In-hospital mortality was higher among patients with a cancer diagnosis, 13.7% vs 9.0% without cancer (**Supplemental Table 1**). Among our cohort, 52.9% had stenting performed early (within 3 days of hospitalization) and 47.1% had stenting performed late. When stenting was performed early, inpatient mortality was 8.9% versus 16.1% when performed late and mean hospital length of stay was 7 versus 19 days (**Supplemental Table 2**). Compared to patients who survived to discharge, patients who died in-hospital were older, had more comorbidities, had more acute organ failures, and more often had stenting performed late (**Supplemental Table 3**).

Trends over time

From 2016 to 2022, the incidence of tracheobronchial stenting during inpatient hospitalizations per 100,000 residents in the US increased from 0.58 to 0.67 per 100,000 residents ($p<0.01$) (**Table 2, Figure 1**), an increase of 16%. The incidence of tracheobronchial stenting peaked in 2021 (0.77 during inpatient hospitalization per 100,000 US residents). Over this same time-period, patients receiving tracheobronchial stenting had more comorbidities over time (4.3 to 4.8; p<0.01) and an increased proportion had a diagnosis of malignancy (65.9% to 74.6%; p<0.01). Likewise, length of stay (11 to 14 days; p<0.01) and in hospital mortality (11.3% to 12.7%; p<0.01) both increased over time (**Supplemental Table 3)**

Regional variation

In 2022, the incidence of tracheobronchial stenting during inpatient hospitalizations was 0.67 per 100,000 residents nationally. However, this varied substantially across geographic regions, ranging from 1.32 per 100,000 in the East North Central Division (Illinois, Indiana, Michigan, Ohio, and Wisconsin) to 0.26 in the Mountain Division (Idaho, Montana, Wyoming, Nevada, Utah, Colorado, Arizona, New Mexico) (**Figure 3**).

State-level analysis

The pooled incidence of tracheobronchial stenting performed during inpatient hospitalization across the six states in 2019 was 0.84 per 100,000 residents, which was slightly higher than the national estimate of 0.76 per 100,000. The combined pooled incidence of tracheobronchial stenting (including those performed in the ambulatory setting and during inpatient hospitalization) was 1.14 per 100,000 residents across the six states in 2019 (**Figure 1, Supplemental Table 2**). The combined incidence of tracheobronchial stenting (inpatient and as an ambulatory procedure) varied across the six states, ranging from 0.57 to 1.60 per 100,000 (**Figure 1**).

Among all tracheobronchial stenting procedures performed in these six states in 2019, 73.3% (95% CI: 68.8%, 76.5%) were performed during inpatient admissions, while 26.7% (95% CI: 23.5%, 30.2%) were performed as ambulatory surgeries. The proportion performed during inpatient hospitalization also varied across states, ranging from 52.6% to 92.6% (**Figure 1**).

Patients receiving tracheobronchial stenting during inpatient hospitalizations were older (median 62 versus 60 years; p=0.02) and critically ill, with 23.8% receiving invasive mechanical ventilation and 33.5% with acute hypoxic respiratory failure (**Supplemental Table 2**). Among hospitalizations with tracheobronchial stenting procedure, in hospital mortality was 11.8%, and 66.5% of patients were discharged to home. Among ambulatory procedures, mortality was 0% and 96.6% of patients were discharged to home (**Supplemental Table 2**).

Discussion

In this nationwide cohort of hospitalized patients in the United States, from 2016 to 2022, the incidence of tracheobronchial stenting was 0.71 per 100,000 residents. Notably, the incidence of tracheobronchial stenting during inpatient hospitalization increased 16% from 2016 to 2022 and alarmingly, across the entire time period, nearly 1 in 8 patients died prior to discharge, which also increased over time.

With the advent of lung cancer screening, there has been a stage shift with a greater proportion of patients diagnosed at an earlier stage and an associated improvement in survival.[31–33] Likewise, with advances in lung cancer treatment including immunotherapy and target treatments, there has been a corresponding improvement in patient outcomes.[34–36] In this cohort, we found increasing tracheobronchial stenting over time, with the majority of patients having a diagnosis of malignancy (which increased over time), and nearly 60% having a specific diagnosis for lung cancer. This trend of increased tracheobronchial stenting in spite of earlier lung cancer diagnosis is perhaps less suprising when considered in the broader landscape. Over this same time period, there has been an increase in Interventional Pulmonology training programs, populating more places in the country with physicians capable and willing to place tracheobronchial stents.[14] Additionally, there has been a proliferation of new self expanding metal stents (SEMS) that are more easily deployed, affording proceduralists an alternative to the traditionally difficult to deploy silicone stent.[6,37–40] While the majority of patients did carry a diagnosis of lung cancer or metastatic cancer, a significant proportion of patients may have had benign disease, with increased incidence of stenting in spite of a block box warning issued by the US Food and Drug Administration (FDA) in 2005 for the placement of SEMS in patients with benign disease due to serious adverse complications.[41–43]

The increase in tracheobronchial stenting follows a well-documented trend in other specialities where a new procedure is followed by increased uptake over time.[44–47] For example, after the introduction of inferior vena cava (IVC) filters, there was a rapid increase and it was only after FDA communication documenting potential serious adverse events was there a decline in use.[46,47]

Our second major finding is that nearly 75% of all tracheobronchial stenting procedures were performed during inpatient hospitalization. However, there is substantial variation across regions and states in the United States, both in terms of the how often tracheobronchial stenting is performed and also where the procedure is performed (inpatient hospitalization versus ambulatory surgery).

The variation between regions and state, while more pronounced, is consistent with other procedures such as the use of bronchoscopy among patients receiving invasive mechanical ventilation, early tracheostomy among mechanically ventilated patients, and the use of coronary angiography, percutaneous coronary intervention, and mechanical circulatory support among patients with acute myocardial infarction related cardiogenic shock.[48–51] While some of this variation may be related to the locations of Interventional Pulmonology training programs and the number of Interventional Pulmonologists practicing within specific regions, the magnitude of variation suggests true differences in practice patterns.[14,42,43,52] Future work is needed to understand how changes in Interventional Pulmonary training programs and increased number of Interventional Pulmonologists impacts the use of tracheobronchial stenting.

The data for tracheobronchial stenting are limited and primarily based on retrospective studies showing improvements in airway patency, dyspnea, and quality of life.[11,15,53–56] While the long-term risk of complications from tracheobronchial stents is high, prior literature has suggested that the risk of peri-operative complications are uncommon.[56–

[58] Yet, we found that nearly 1 in 8 patients undergoing tracheobronchial stenting as an inpatient died prior to hospital discharge. Notably, mortality among patients receiving this as an ambulatory procedure was 0%, highlighting that for patients well enough to be in the ambulatory setting, tracheobronchial stenting is a very safe procedure; however, inpatients undergoing tracheobronchial stenting are a distinct patient population. Discussions with hospitalized patients regarding the risks/benefits of tracheobronchial stenting should account for these differences in outcomes.

Based on our data source, we cannot determine if mortality was directly related to the tracheobronchial procedure. Yet, the fact that inpatient mortality was consistent across both our national and state-level data source suggests that there is high inpatient mortality in this patient population receiving tracheobronchial stenting, which is not described in the primarily small, single center retrospective cohort studies describing the safety of tracheobronchial stenting.

Based on the current population of the United States, our data suggest that each year nearly 300 patients receiving tracheobronchial stenting during an inpatient hospitalization die prior to hospital discharge.[59] This high inpatient mortality for a palliative intervention is highly concerning and highlights the need to better identify which patients receiving tracheobronchial stenting are most likely to benefit, and potentially more importantly, which patients will not benefit and are at high risk of peri-procedural mortality. It also provides a data point to provide informed consent to patients, highlighting that this intervention is palliative and still associated with a high in-hospital mortality.

Limitations

While this project has several strengths, there are also several limitations. First, we used procedural coding to capture our outcome of interest. However, prior work has shown high specificity for ICD codes to identify procedures.[60] Second, based on our data source, we are unable to assess the impact of tracheobronchial stenting on clinical outcomes for any individual patient nor can we determine the exact etiology for stenting. However, outcomes were consistent in analyses among patients with and without a diagnosis of malignancy. Third, while we attempted to capture a geographically diverse set of states across the United States, it is possible that these states are not representative. However, the pooled incidence of tracheobronchial stenting performed during hospitalizations across the six states was consistent with the national estimate, suggesting that these states may be representative.

Conclusion

From 2016 to 2022, the use of tracheobronchial stenting in the United States increased and varied markedly across states and regions. For procedures performed during inpatient hospitalization, nearly 1 in 8 died prior to hospital discharge. This variation and high mortality highlights the need to clarify which patients most benefit from tracheobronchial stenting.

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Table 1: Hospitalizations receiving tracheobronchial stenting in the US (2016-2022)	
Incidence per 100,000 (95% CI)	0.71 (0.66, 0.76)
Demographics	
Female, %	49.5
Age (mean in years)	62
Race, %	
White	68.1
Black	15.0
Other	8.6
Hispanic	8.3
Hospital Characteristics, %	
Location and teaching status	
Urban teaching	43.1
Urban non teaching	30.1
Rural	26.8
Hospital Size	
Large	33.5
Medium	22.9
Small	43.5
Hospital Ownership	
Private, nonprofit	71.2
Government, nonfederal	14.9
Private, investor-owned, for profit	13.9
Clinical characteristics	
Elixhauser comorbidities, mean	4.7
Malignancy (any), %	69.7
Lung cancer, %	59.1%
Metastatic cancer, %	40.7%
Lymphoma	2.8%
Number of acute organ dysfunctions, mean	0.8
Timing of stent placement	
Early (within 3 days), %	52.9%
Late (after 3 days), %	47.1%
Mean length of stay, days	13
Discharged home[1], %	69.6
In-hospital mortality, %	12.3

[1]Includes discharged home, discharged home with health services, and left against medical advice
CI: confidence interval

Table 2: Trends in tracheobronchial stenting performed during inpatient hospitalization (2016-2022)	2016	2018	2020	2022	p-value
Overall incidence per 100,000 residents[1] (95% CI)	0.58 (0.47, 0.69)	0.76 (0.61, 0.91)	0.69 (0.57, 0.82)	0.67 (0.55, 0.79)	<0.01
Patient characteristics					
Demographics					
Female	52.7	52.0	53.3	49.0	0.57
Age	62	63	61	62	0.38
Race					
White	68.6	71.9	63.6	69.3	0.93
Black	14.3	15.0	18.2	14.0	
Other	10.0	5.8	10.1	7.1	
Hispanic	7.1	7.3	8.1	9.6	
Primary payer					
Medicare	43.7	54.8	46.4	48.3	0.14
Private Insurance	35.4	24.9	30.7	28.5	
Medicaid	16.4	15.9	14.4	16.7	
Other (self-pay, no charge, other)	4.6	4.2	8.5	6.5	
Clinical characteristics					
Elixhauser comorbidities, median	4.3	4.7	4.8	4.8	<0.01
Comorbidities					
Malignancy	65.9	65.5	73.5	74.6	<0.01
Lung transplant	8.0	8.0	5.2	5.8	0.03
Acute organ dysfunction, %					
Respiratory	26.1	25.9	24.1	21.4	0.18
Cardiovascular	17.1	20.3	18.0	18.9	0.68
Renal	13.3	14.9	17.6	16.9	0.54
Hematologic	6.4	7.4	6.3	8.2	0.54
Neurologic	5.3	10.4	8.0	10.5	<0.01
Total number of organ dysfunctions	0.7	0.8	0.7	0.8	0.50
Timing of stent placement					
Early (within 3 days), %	56.2%	57.6%	49.3%	50.8%	<0.01
Late (after 3 days), %	43.7%	42.4%	50.7%	49.2%	
Length of stay in days, mean	11	12	13	14	<0.01

Discharged to home[2]	70.4	69.1	71.3	71.0	0.35
In-hospital mortality	11.3	11.2	11.5	12.7	<0.001

[1]Rate per 100,000 residents calculated using the United States Census bureau estimate for population in that year
[2]Includes discharged home, discharged home with health services, and left against medical advice
CI: confidence interval; IQR: interquartile range

Figure 1: Tracheobronchial stenting trends in the United States (2016 to 2022)

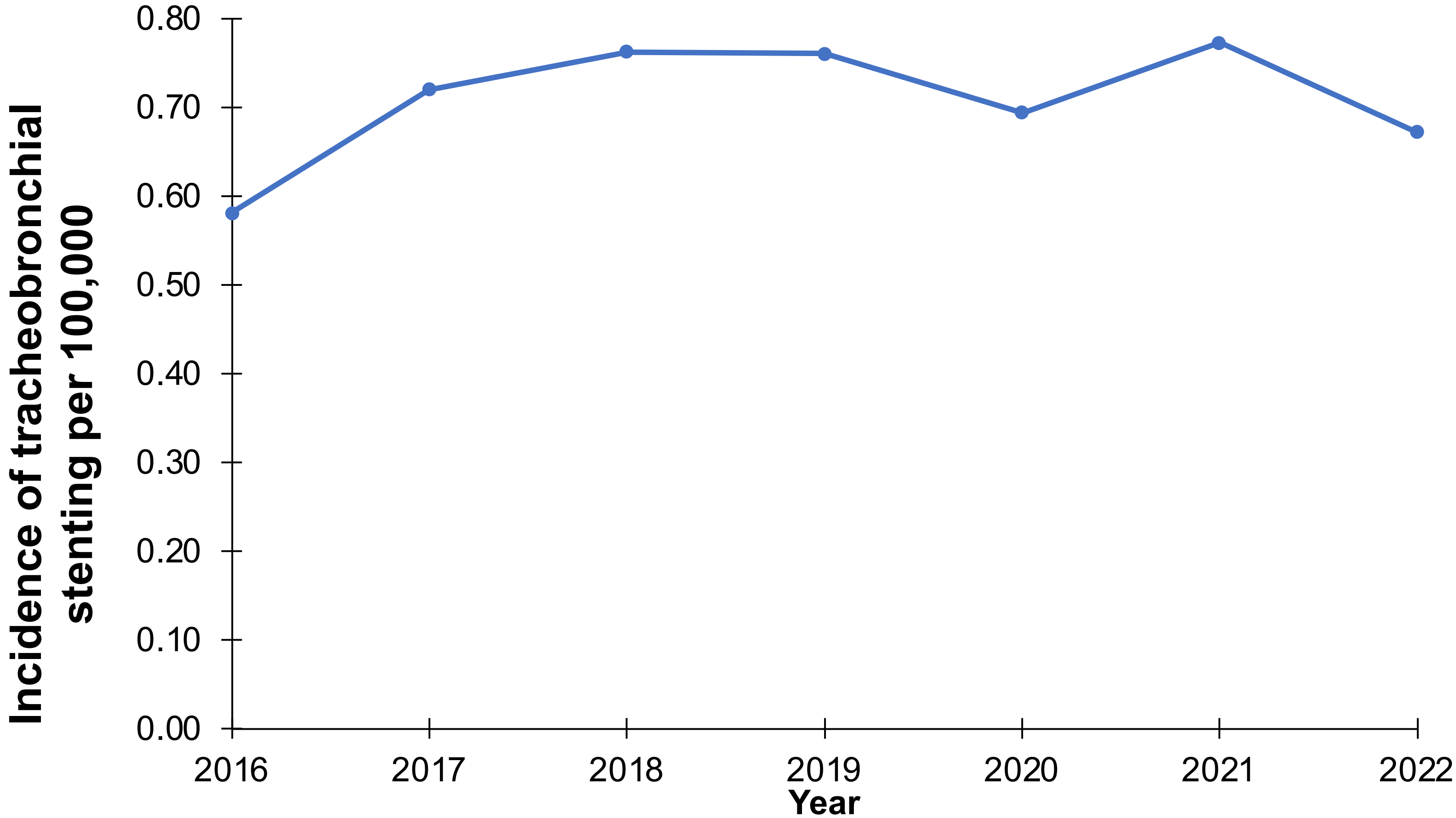
Figure 1 Legend: Line graph showing temportal trends in the use of tracheobronchial stenting in the United States from 2016 to 2022. The black line represents the incidence of tracheobronchial stenting during inpatient hospitalizations per 100,000 residents. The blue line represents estimated incidence of tracheobronchial stenting performed as an ambulatory procedure per 100,000 residents with confidence interval representing ranges seen across individual states in 2019. The red line represents the combined incidence of tracheobronchial stenting performed during inpatient hospitalization and as ambulatory procedure per 100,000.

Figure 2: Tracheobronchial stenting across regions of the United States in 2022

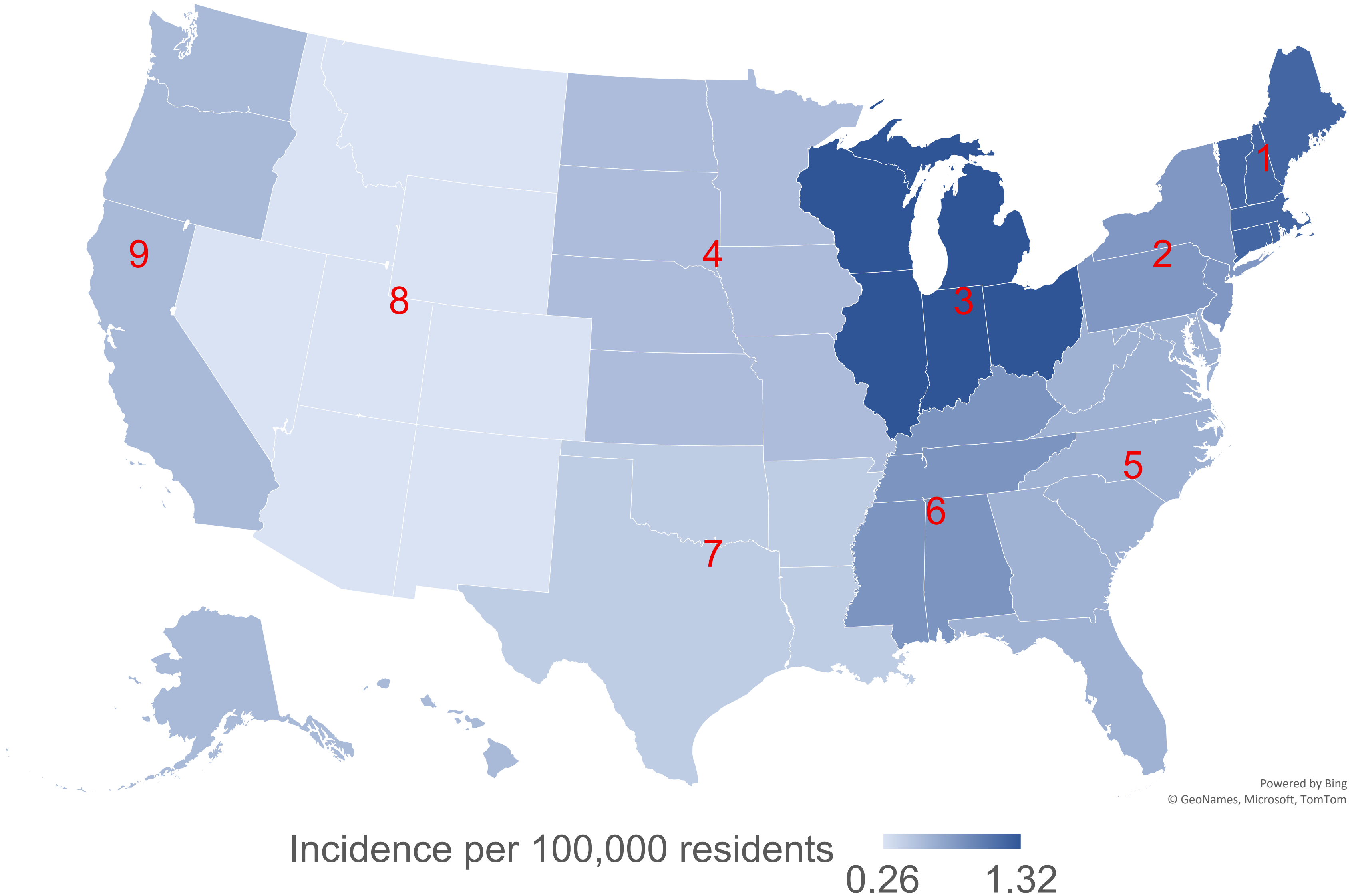
Figure 2 Legend: Map showing the incidence of hospitalizations receiving tracheobronchial stenting across regions defined by US Census division in 2022. US Census division tracts: 1: Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont; 2: New Jersey, New York, and Pennsylvania; 3: Illinois, Indiana, Michigan, Ohio, and Wisconsin; 4: Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, and South Dakota; 5: Delaware, District of Columbia, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, and West Virginia; 6: Alabama, Kentucky, Mississippi, and Tennessee; 7: Arkansas, Louisiana, Oklahoma, and Texas; 8: Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, and Wyoming; 9: Alaska, California, Hawaii, Oregon, Washington

Figure 3: Tracheobronchial stenting overall and by state in 2019 per 100,000 residents

Figure 3 Legend: Stacked bar graph showing the incidence of tracheobronchial stenting per 100,000 residents, overall and across states. The national estimate, pooled overall state incidence, and each individual state are on the y-axis. The incidence per 100,000 is on the x-axis. The darker blue color indicates tracheobronchial stenting performed as an inpatient procedure and the lighter blue represents tracheobronchial stenting performed as an ambulatory procedure. The number to the right of each line represents the total incidence. The national estimate is calculated shows the incidence of tracheobronchial stenting per 100,000 residents within the United States in 2019 using the National Inpatient Sample. The overall incidence shows the pooled estimate across 6 states within the United States in 2019.



Incidence of tracheobronchial stenting per 100,000
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
2016
2017
2018
2019
2020
2021
2022
Year


1
2
3
4
5
6
7
8
9
Powered by Bing
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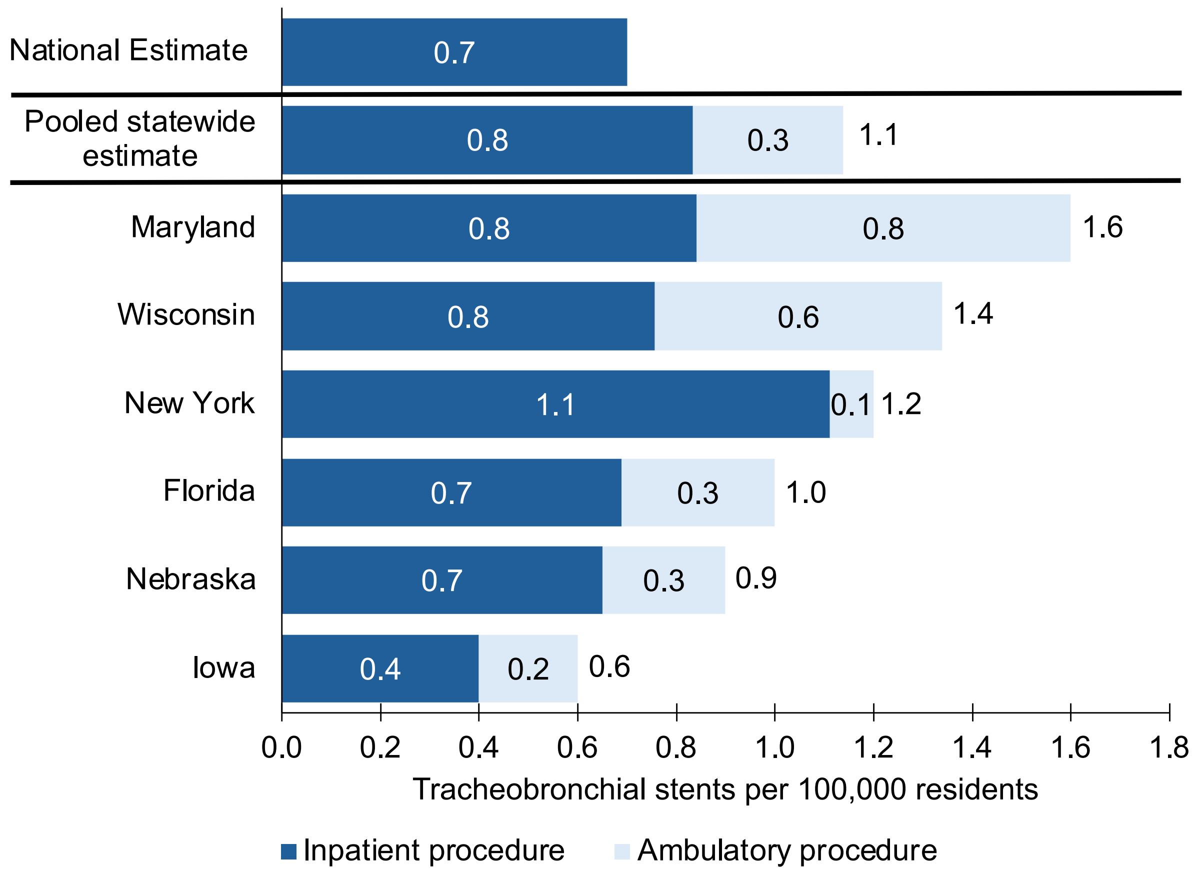
Incidence per 100,000 residents
0.26
1.32


National Estimate
0.7
Pooled statewide estimate
0.8
0.3
1.1
Maryland
0.8
0.8
1.6
Wisconsin
0.8
0.6
1.4
New York
1.1
0.1
1.2
Florida
0.7
0.3
1.0
Nebraska
0.7
0.3
0.9
Iowa
0.4
0.2
0.6
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Tracheobronchial stents per 100,000 residents
Inpatient procedure
Ambulatory procedure